Grade 11 — Learning Evidence 3
C8, C9 and C10 — Exam Solutions — Version 2

1st Term 2026–2027

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Learning objective

Apply continuous probability models and sampling distributions to investigate economic and financial questions, justify statistical methods, and communicate evidence-based conclusions.

Criteria assessment

Partial credit may be awarded when the correct procedure is clearly demonstrated, even if arithmetic errors are made. No points will be awarded for answers without a demonstrated procedure.

A minimum of 4 points is required to pass each criterion. A 1-point penalty will be applied to each criterion if the student's name is not written on the assessment or if the assessment is submitted late.

№	C8
1–5	5
6–10	5
Total	**10**

№	C9
11–13	6
14–16	6
Total	**12**

The worked solutions show the reasoning using the supplied normal and Student's t reference sheets; no interpolation, CDF calculator, or inverse-CDF calculator is needed. C8 is worth **10 points**, C9 **12 points**, and C10 **6 points**: **28 points total**. At least **4 points per criterion** are required to pass that criterion. All calculations within one problem share its stated allocation, including both interval endpoints.

Evaluated criteria

C8 Uses Student's t-distribution tables, degrees of freedom, and one-tailed or two-tailed areas to determine critical values, central probabilities, and tail probabilities or probability ranges.

C9 Calculates and compares critical values and probability areas from the standard normal and Student's t-distributions, selecting and justifying the appropriate distribution according to whether the population standard deviation is known or estimated from the sample.

C10 Integrates continuous probability modelling, simulation, sampling analysis, and portfolio-risk measures to investigate an economic or financial question, justifying the selected methods and communicating results, assumptions, limitations, and evidence-based conclusions.

Models and table conventions

Problems 1–10 use the supplied signed-deviation model $X = m + aT$, $T \sim t_\nu$. Problems 11–16 compare

$$X_T = m + aT, \quad T \sim t_\nu, \qquad X_N = m + aZ, \quad Z \sim N(0,1), \qquad \nu = n - 1 .$$

Here n is the number of sample observations, m is the location (and mean for $\nu > 1$), and $a > 0$ is the scale. Use the stated *approximate standard deviation* directly as a, without adjustment. This is a teaching approximation:

$$\mathrm{SD}(X_T) = a\sqrt{\frac{\nu}{\nu - 2}} \quad (\nu > 2), \qquad \mathrm{SD}(X_N) = a.$$

The two models share a scale, not an actual variance. The approximation is coarser for small ν and improves as ν increases. Negative deviations mean below the contextual reference (early or under budget); positive deviations mean above it (late or over budget). Inverse intervals enclose modelled deviations. Standardize and back-transform using

$$c = \frac{x - m}{a}, \qquad x = m + ac .$$

The year-local sheets are in `book2026-2027/fundamentals/normal-table.tex` (and its PDF). In the t table, row ν and one-tail column α give positive t^* satisfying

$$P(T > t^*) = \alpha, \qquad P(T < -t^*) = \alpha, \qquad P(-t^* < T < t^*) = 1 - 2\alpha.$$

Obtain negative cutoffs by symmetry. The printed columns are $0.30, 0.20, 0.10, 0.05, 0.025, 0.01, 0.005$. Direct bounds recover printed three-decimal entries exactly through standardization. Report C8 probabilities as implied by these entries, and contextual cutoffs to two decimals.

The normal table gives four-decimal cumulative areas $\Phi(z) = P(Z \le z)$ for positive z. Use $\Phi(-z) = 1 - \Phi(z)$. For direct normal probabilities, round each standardized boundary to two decimals *before* lookup. For inverse normal problems, select the closest printed CDF entry. At targets 0.0500 and 0.9500, magnitudes 1.64 and 1.65 tie; choose $|z| = 1.64$, as in the C9 activity. Use the CDF table for inverse normal lookups.

For **every C9 problem**, calculate both results and

$$E_A = |r_T - r_N|, \qquad E_{Pt} = \frac{100E_A}{|r_T|}, \qquad E_{Pn} = \frac{100E_A}{|r_N|}.$$

For direct problems, r_T, r_N are probabilities; for inverse problems, they are contextual cutoffs after back-transformation. Compare inverse interval endpoints separately. A zero denominator makes that percentage undefined; the data below avoid zero denominators. These measures compare model results and do not establish error against observed truth. Report C9 probabilities and their E_A to four decimals; contextual cutoffs and their E_A to two decimals with units; and percentage comparisons to two decimals. Use unrounded back-transformed cutoffs for inverse comparisons, rounding only final results.

All results are table-based approximations, not exact distribution values: printed critical values and CDF entries are rounded. Equalities in the worked arithmetic use selected table values as working values. Both models are continuous, so endpoints have probability zero. The formally stated criteria above are retained; the supplied activity models govern the C8/C9 calculations here.

1 Warehouse energy deviations [C8-1]

C8 – Direct interval above the mean. Over $n = 18$ days, electricity-use deviation from forecast, in kWh, has mean $m = 2.4$ and approximate standard deviation $a = 7$. Use $X = 2.4 + 7T$, $T \sim t_{n-1}$.

Questions/Tasks

C8 Problem 1. Find $P(8.441 < X < 14.58)$. **[1]**

C8 – Problem 1: Direct interval above the mean solution [1]

Here $\nu = 18 - 1 = 17$. Both bounds exceed $m = 2.4$:

$$c_L = \frac{8.441 - 2.4}{7} = 0.863, \qquad c_U = \frac{14.58 - 2.4}{7} = 1.740.$$

Row 17, columns $\alpha = 0.20$ and 0.05, gives 0.863 and 1.740. Subtract the nested right tails:

$$P(8.441 < X < 14.58) \approx 0.20 - 0.05 = \boxed{0.15}.$$

About 15% of days use between 8.44 and 14.58 kWh above forecast.

2 Retail revenue alerts [C8-1]

C8 – Direct unequal combined tails. Over $n = 8$ weeks, revenue deviation from plan, in thousands of dollars, has mean $m = 1.6$ and approximate standard deviation $a = 5$. Use $X = 1.6 + 5T$, $T \sim t_{n-1}$.

Questions/Tasks

C8 Problem 2. Find the probability of an alert: $X < -7.875$ or $X > 8.675$. **[1]**

C8 – Problem 2: Direct unequal combined tails solution [1]

Here $\nu = 8 - 1 = 7$ and

$$c_L = \frac{-7.875 - 1.6}{5} = -1.895, \qquad c_U = \frac{8.675 - 1.6}{5} = 1.415.$$

Row 7, columns $\alpha = 0.05$ and 0.10, gives 1.895 and 1.415. Add the disjoint tails:

$$P(X < -7.875 \text{ or } X > 8.675) \approx 0.05 + 0.10 = \boxed{0.15}.$$

About 15% of weeks trigger an alert: revenue is more than 7.88 thousand dollars below plan or 8.68 thousand dollars above it.

3 Courier arrival deviations [C8-1]

C8 – Direct left tail. A sample of $n = 11$ courier trips has arrival-time deviation from schedule, in minutes, with mean $m = 1.2$ and approximate standard deviation $a = 3.5$. Use $X = 1.2 + 3.5T$, $T \sim t_{n-1}$.

Questions/Tasks

C8 Problem 3. Find $P(X < -5.142)$. **[1]**

C8 – Problem 3: Direct left tail solution [1]

Here $\nu = 11 - 1 = 10$. Standardize:

$$c = \frac{-5.142 - 1.2}{3.5} = -1.812.$$

Row 10, column $\alpha = 0.05$, gives $t^* = 1.812$. By symmetry,

$$P(X < -5.142) = P(T < -1.812) \approx \boxed{0.05}.$$

About 5% of trips arrive more than 5.14 minutes early.

4 Freight cost deviations [C8-1]

C8 – Direct right tail. A sample of $n = 14$ consignments has cost deviation from budget, in dollars, with mean $m = -5$ and approximate standard deviation $a = 9$. Use $X = -5 + 9T$, $T \sim t_{n-1}$.

Questions/Tasks

C8 Problem 4. Find $P(X > -12.83)$. **[1]**

C8 – Problem 4: Direct right tail solution [1]

Here $\nu = 14 - 1 = 13$ and

$$c = \frac{-12.83 - (-5)}{9} = -0.870.$$

Row 13, column $\alpha = 0.20$, gives $t^* = 0.870$. Symmetry gives $P(T < -0.870) \approx 0.20$, so

$$P(X > -12.83) = 1 - P(T < -0.870) \approx 1 - 0.20 = \boxed{0.80}.$$

About 80% have costs above the threshold of approximately \$12.83 below budget. The cutoff is below $m = -5$, so the right-tail event includes more than half the symmetric model.

5 Crop shipment deviations [C8-1]

C8 – Direct interval below the mean. A sample of $n = 9$ crop shipments has mass deviation from forecast, in tonnes, with mean $m = -1.2$ and approximate standard deviation $a = 3.6$. Use $X = -1.2 + 3.6T$, $T \sim t_{n-1}$.

Questions/Tasks

C8 Problem 5. Find $P(-9.5016 < X < -4.4004)$. **[1]**

C8 – Problem 5: Direct interval below the mean solution [1]

Here $\nu = 9 - 1 = 8$. Both bounds lie below $m = -1.2$:

$$c_L = \frac{-9.5016 - (-1.2)}{3.6} = -2.306, \qquad c_U = \frac{-4.4004 - (-1.2)}{3.6} = -0.889.$$

Row 8, columns $\alpha = 0.025$ and 0.20, gives 2.306 and 0.889. By symmetry the cumulative areas at the negative bounds are 0.025 and 0.20:

$$P(-9.5016 < X < -4.4004) \approx 0.20 - 0.025 = \boxed{0.175}.$$

About 17.5% of shipments fall short of forecast by between 4.40 and 9.50 tonnes.

6 Package mass alerts [C8-1]

C8 – Inverse equal-tail alert limits. A sample of $n = 10$ packages has mass deviation from target, in grams, with mean $m = -0.5$ and approximate standard deviation $a = 1.4$. Use $X = -0.5 + 1.4T$, $T \sim t_{n-1}$.

Questions/Tasks

C8 Problem 6. Find two alert limits leaving a combined outside probability of 0.05, split equally. **[1]**

C8 – Problem 6: Inverse equal-tail alert limits solution [1]

Here $\nu = 10 - 1 = 9$. Each tail has probability $0.05/2 = 0.025$. Row 9, column $\alpha = 0.025$, gives $t^* = 2.262$, so $(c_L, c_U) = (-2.262, 2.262)$:

$$L = -0.5 - 1.4(2.262) = -3.6668 \approx \boxed{-3.67 \text{ g}},$$

$$U = -0.5 + 1.4(2.262) = 2.6668 \approx \boxed{2.67 \text{ g}}.$$

About 2.5% of packages fall below L and 2.5% above U. An alert means more than 3.67 g under target or 2.67 g over target.

7 Operating cost cutoff [C8-1]

C8 – Inverse lower-tail cutoff. Over $n = 16$ production days, operating-cost deviation from budget, in dollars, has mean $m = -3.5$ and approximate standard deviation $a = 6$. Use $X = -3.5 + 6T$, $T \sim t_{n-1}$.

Questions/Tasks

C8 Problem 7. Find x such that $P(X < x) = 0.10$. **[1]**

C8 – Problem 7: Inverse lower-tail cutoff solution [1]

Here $\nu = 16 - 1 = 15$. The lower-tail target is 0.10. Row 15, column $\alpha = 0.10$, gives $t^* = 1.341$. By symmetry $c = -1.341$; back-transform:

$$x = -3.5 + 6(-1.341) = -11.546 \approx \boxed{-11.55 \text{ dollars}}.$$

About 10% of days have costs more than \$11.55 below budget.

8 Dispatch time alerts [C8-1]

C8 – Inverse unequal-tail alert limits. A sample of $n = 17$ dispatches has departure-time deviation from schedule, in minutes, with mean $m = -2.5$ and approximate standard deviation $a = 7$. Use $X = -2.5 + 7T$, $T \sim t_{n-1}$.

Questions/Tasks

C8 Problem 8. Find L, U with $P(X < L) = 0.01$ and $P(X > U) = 0.10$. **[1]**

C8 – Problem 8: Inverse unequal-tail alert limits solution [1]

Here $\nu = 17 - 1 = 16$. Use the tails separately. Row 16, column $\alpha = 0.01$, gives 2.583, so $c_L = -2.583$ by symmetry. Column $\alpha = 0.10$ gives $c_U = 1.337$:

$$L = -2.5 + 7(-2.583) = -20.581 \approx \boxed{-20.58 \text{ minutes}},$$

$$U = -2.5 + 7(1.337) = 6.859 \approx \boxed{6.86 \text{ minutes}}.$$

About 1% depart more than 20.58 minutes early, and 10% more than 6.86 minutes late. The combined alert probability is $0.01 + 0.10 = 0.11$; 0.89 lies between the limits.

9 Loading time cutoff [C8-1]

C8 – Inverse upper-tail cutoff. A sample of $n = 22$ loading jobs has duration deviation from target, in minutes, with mean $m = 2.5$ and approximate standard deviation $a = 8$. Use $X = 2.5 + 8T$, $T \sim t_{n-1}$.

Questions/Tasks

C8 Problem 9. Find x such that $P(X > x) = 0.025$. **[1]**

C8 – Problem 9: Inverse upper-tail cutoff solution [1]

Here $\nu = 22 - 1 = 21$. Row 21, column $\alpha = 0.025$, gives $c = 2.080$:

$$P(T > 2.080) \approx 0.025, \qquad x = 2.5 + 8(2.080) = 19.140 \approx \boxed{19.14 \text{ minutes}}.$$

About 2.5% of jobs take more than 19.14 minutes longer than target.

10 Demand deviation interval [C8-1]

C8 – Inverse central interval. Over $n = 13$ weeks, demand deviation from forecast, in tonnes, has mean $m = 5$ and approximate standard deviation $a = 10$. Use $X = 5 + 10T$, $T \sim t_{n-1}$.

Questions/Tasks

C8 Problem 10. Find $L < U$ enclosing the central 90% of deviations with equal excluded tails. **[1]**

C8 – Problem 10: Inverse central interval solution [1]

Here $\nu = 13 - 1 = 12$. Each tail is $(1 - 0.90)/2 = 0.05$. Row 12, column $\alpha = 0.05$, gives $t^* = 1.782$, so $c_L = -1.782$, $c_U = 1.782$:

$$L = 5 - 10(1.782) = -12.820 \approx \boxed{-12.82 \text{ tonnes}},$$

$$U = 5 + 10(1.782) = 22.820 \approx \boxed{22.82 \text{ tonnes}}.$$

About 90% lie between 12.82 tonnes below forecast and 22.82 tonnes above it, with 5% in each excluded tail.

11 Delivery schedule deviations [C9-2]

C9 – Direct right tail. A sample of $n = 13$ deliveries has arrival-time deviation from schedule, in minutes, with mean $m = 3.2$ and approximate standard deviation $a = 4.5$. Compare $X_T = 3.2 + 4.5T$, $T \sim t_{n-1}$, with $X_N = 3.2 + 4.5Z$.

Questions/Tasks

C9 Problem 11. Find $P(X > -0.7285)$. Find both model results, calculate E_A, E_{Pt}, and E_{Pn}. **[2]**

C9 – Problem 11: Direct right tail solution [2]

Here $\nu = 13 - 1 = 12$; $c = (-0.7285 - 3.2)/4.5 = -0.873$. Row 12, column $\alpha = 0.20$, gives $t^* = 0.873$:

$$p_T = 1 - P(T < -0.873) \approx 1 - 0.2000 = \boxed{0.8000}.$$

For normal, round to $z = -0.87$. Row 0.8, column 0.07, gives $\Phi(0.87) = 0.8078$:

$$\Phi(-0.87) = 1 - 0.8078 = 0.1922, \qquad p_N = 1 - 0.1922 = \boxed{0.8078}.$$

$$E_A = |0.8000 - 0.8078| = \boxed{0.0078},$$

$$E_{Pt} = \frac{100(0.0078)}{|0.8000|} \approx \boxed{0.98\%},$$

$$E_{Pn} = \frac{100(0.0078)}{|0.8078|} \approx \boxed{0.97\%}.$$

The event includes arrivals less than about 0.73 minutes early and all late arrivals. Normal assigns 80.78%, which is 0.78 percentage points higher than t's 80.00%. Both exceed 0.5 because the cutoff is below $m = 3.2$; t does not give the larger probability for this event.

12 Fuel forecast deviations [C9-2]

C9 – Direct interval crossing the mean. Over $n = 14$ operating days, fuel-use deviation from forecast, in litres, has mean $m = 1.2$ and approximate standard deviation $a = 7$. Compare $X_T = 1.2 + 7T$, $T \sim t_{n-1}$, with $X_N = 1.2 + 7Z$.

Questions/Tasks

C9 Problem 12. Find $P(-8.25 < X < 16.32)$. Find both model results, calculate E_A, E_{Pt}, and E_{Pn}. **[2]**

C9 – Problem 12: Direct interval crossing the mean solution [2]

Here $\nu = 14 - 1 = 13$. Standardize:

$$c_L = \frac{-8.25 - 1.2}{7} = -1.350, \qquad c_U = \frac{16.32 - 1.2}{7} = 2.160.$$

Row 13, columns $\alpha = 0.10$ and 0.025, gives 1.350 and 2.160. The interval crosses $m = 1.2$ and excludes unequal tails:

$$p_T \approx 1 - 0.1000 - 0.0250 = \boxed{0.8750}.$$

For normal, round to $z_L = -1.35$ and $z_U = 2.16$. Row 1.3, column 0.05, gives $\Phi(1.35) = 0.9115$; row 2.1, column 0.06, gives $\Phi(2.16) = 0.9846$:

$$p_N = 0.9846 - (1 - 0.9115) = 0.9846 - 0.0885 = \boxed{0.8961}.$$

$$E_A = |0.8750 - 0.8961| = \boxed{0.0211},$$

$$E_{Pt} = \frac{100(0.0211)}{|0.8750|} \approx \boxed{2.41\%},$$

$$E_{Pn} = \frac{100(0.0211)}{|0.8961|} \approx \boxed{2.35\%}.$$

Fuel use lies from 8.25 litres below forecast to 16.32 litres above it with probability about 87.50% under t or 89.61% under normal. Normal assigns 2.11 percentage points more to this interval.

13 Sales revenue shortfalls [C9-2]

C9 – Direct left tail. Over $n = 11$ weeks, sales-revenue deviation from target, in thousands of dollars, has mean $m = -2.5$ and approximate standard deviation $a = 6$. Compare $X_T = -2.5 + 6T$, $T \sim t_{n-1}$, with $X_N = -2.5 + 6Z$.

Questions/Tasks

C9 Problem 13. Find $P(X < -13.372)$. Find both model results, calculate E_A, E_{Pt}, and E_{Pn}. **[2]**

C9 – Problem 13: Direct left tail solution [2]

Here $\nu = 11 - 1 = 10$; $c = (-13.372 - (-2.5))/6 = -1.812$. Row 10, column $\alpha = 0.05$, gives $t^* = 1.812$; symmetry gives

$$p_T = P(T < -1.812) \approx \boxed{0.0500}.$$

For normal, round to $z = -1.81$. Row 1.8, column 0.01, gives $\Phi(1.81) = 0.9649$:

$$p_N = 1 - 0.9649 = \boxed{0.0351}.$$

$$E_A = |0.0500 - 0.0351| = \boxed{0.0149},$$
$$E_{Pt} = \frac{100(0.0149)}{|0.0500|} \approx \boxed{29.80\%},$$
$$E_{Pn} = \frac{100(0.0149)}{|0.0351|} \approx \boxed{42.45\%}.$$

A shortfall greater than 13.37 thousand dollars has probability about 5.00% under t, compared with 3.51% under normal, a difference of 1.49 percentage points.

14 Cargo mass deviation interval [C9-2]

C9 – Inverse central interval. A sample of $n = 11$ cargo loads has mass deviation from target, in kg, with mean $m = 1.8$ and approximate standard deviation $a = 5$. Compare $X_T = 1.8 + 5T$, $T \sim t_{n-1}$, with $X_N = 1.8 + 5Z$.

Questions/Tasks

C9 Problem 14. Find the central 95% interval with equal excluded tails under each model. Calculate E_A, E_{Pt}, and E_{Pn} separately for each endpoint. **[2]**

C9 – Problem 14: Inverse central interval solution [2]

Here $\nu = 11 - 1 = 10$. Each tail is $(1 - 0.95)/2 = 0.025$, so cumulative targets are 0.0250 and 0.9750. Standardize unknown limits with $c_L = (L - 1.8)/5$ and $c_U = (U - 1.8)/5$. Row 10, column $\alpha = 0.025$, gives $t^* = 2.228$, hence $(c_{T,L}, c_{T,U}) = (-2.228, 2.228)$. For normal, row 1.9, column 0.06, gives $\Phi(1.96) = 0.9750$ exactly at printed precision; symmetry gives $\Phi(-1.96) = 0.0250$. Thus $(c_{N,L}, c_{N,U}) = (-1.96, 1.96)$.

Back-transform, retaining unrounded values:

$$L_T = 1.8 - 5(2.228) = -9.340 \approx \boxed{-9.34 \text{ kg}},$$
$$U_T = 1.8 + 5(2.228) = 12.940 \approx \boxed{12.94 \text{ kg}},$$
$$L_N = 1.8 - 5(1.96) = -8.00 = \boxed{-8.00 \text{ kg}},$$
$$U_N = 1.8 + 5(1.96) = 11.60 = \boxed{11.60 \text{ kg}}.$$

Lower endpoint comparison:

$$E_{A,L} = |-9.340 - (-8.00)| = 1.340 \approx \boxed{1.34 \text{ kg}},$$
$$E_{Pt,L} = \frac{100(1.340)}{|-9.340|} \approx \boxed{14.35\%},$$
$$E_{Pn,L} = \frac{100(1.340)}{|-8.00|} \approx \boxed{16.75\%}.$$

Upper endpoint comparison:

$$E_{A,U} = |12.940 - 11.60| = 1.340 \approx \boxed{1.34 \text{ kg}},$$
$$E_{Pt,U} = \frac{100(1.340)}{|12.940|} \approx \boxed{10.36\%},$$
$$E_{Pn,U} = \frac{100(1.340)}{|11.60|} \approx \boxed{11.55\%}.$$

The central intervals are approximately $(-9.34, 12.94)$ kg under t and $(-8.00, 11.60)$ kg under normal, with 2.5% in each excluded tail. Negative limits mean below target mass; positive limits mean above it. The normal interval extends about 1.34 kg less far from m at each end. Percentage comparisons differ because the endpoint magnitudes differ.

15 Production cost savings [C9-2]

C9 – Inverse lower-tail cutoff. A sample of $n = 9$ batches has cost deviation from budget, in dollars, with mean $m = -1.3$ and approximate standard deviation $a = 3$. Compare $X_T = -1.3 + 3T$, $T \sim t_{n-1}$, with $X_N = -1.3 + 3Z$.

Questions/Tasks

C9 Problem 15. Find the cutoff x below which 5% of deviations fall. Find both model results, calculate E_A, E_{Pt}, and E_{Pn}. **[2]**

C9 – Problem 15: Inverse lower-tail cutoff solution [2]

Here $\nu = 9 - 1 = 8$; the cumulative target is 0.0500, with $c = (x + 1.3)/3$. Row 8, column $\alpha = 0.05$, gives $t^* = 1.860$, so $c_T = -1.860$. For normal, row 1.6, columns 0.04 and 0.05, gives 0.9495 and 0.9505. The negative cumulative entries are 0.0505 and 0.0495, equally close to 0.0500. The tie rule selects $c_N = -1.64$.

$$x_T = -1.3 + 3(-1.860) = -6.8800 \approx \boxed{-6.88 \text{ dollars}},$$
$$x_N = -1.3 + 3(-1.64) = -6.22 = \boxed{-6.22 \text{ dollars}}.$$

Use unrounded contextual results:

$$E_A = |-6.8800 - (-6.22)| = 0.6600 \approx \boxed{0.66 \text{ dollars}},$$
$$E_{Pt} = \frac{100(0.6600)}{|-6.8800|} \approx \boxed{9.59\%},$$
$$E_{Pn} = \frac{100(0.6600)}{|-6.22|} \approx \boxed{10.61\%}.$$

About 5% of batches save more than \$6.88 under t, or \$6.22 under normal. The normal threshold is about \$0.66 closer to the mean deviation.

16 Order processing cutoff [C9-2]

C9 – Inverse upper-tail cutoff. A sample of $n = 18$ orders has processing-time deviation from target, in minutes, with mean $m = 2.4$ and approximate standard deviation $a = 4$. Compare $X_T = 2.4 + 4T$, $T \sim t_{n-1}$, with $X_N = 2.4 + 4Z$.

Questions/Tasks

C9 Problem 16. Find the cutoff x exceeded by 10% of deviations. Find both model results, calculate E_A, E_{Pt}, and E_{Pn}. **[2]**

C9 – Problem 16: Inverse upper-tail cutoff solution [2]

Here $\nu = 18 - 1 = 17$; convert right tail 0.10 to cumulative target 0.9000. With $c = (x - 2.4)/4$, row 17, column $\alpha = 0.10$, gives $c_T = 1.333$. For normal, row 1.2, column 0.08, gives $\Phi(1.28) = 0.8997$, closer to 0.9000 than $\Phi(1.29) = 0.9015$ (column 0.09). Thus $c_N = 1.28$.

$$x_T = 2.4 + 4(1.333) = 7.7320 \approx \boxed{7.73 \text{ minutes}},$$
$$x_N = 2.4 + 4(1.28) = 7.52 = \boxed{7.52 \text{ minutes}}.$$

Use unrounded cutoffs:

$$E_A = |7.7320 - 7.52| = 0.2120 \approx \boxed{0.21 \text{ minutes}},$$
$$E_{Pt} = \frac{100(0.2120)}{|7.7320|} \approx \boxed{2.74\%},$$
$$E_{Pn} = \frac{100(0.2120)}{|7.52|} \approx \boxed{2.82\%}.$$

About 10% of orders take more than 7.73 minutes above target under t, or 7.52 minutes under normal. The normal cutoff is about 0.21 minutes lower.

17 Research option [C10-6]

Exam:

Questions/Tasks

C10 Select one model and describe plausible real-life data in 1–2 sentences. **[6]**

(a) Linear probability density over a finite interval.
(b) Exponential distribution (waiting times).
(c) Normal distribution (measurements).

Online report: Submit a PDF through **Open** using real economic or financial data and the selected model. Include simulation, sampling, portfolio risk, methods, assumptions, limitations, and evidence-based conclusions. The report is the C10 evidence.

C10 – Acceptable alternative brief responses [6]

This selection task has no unique answer. For this worked response, select **(b)** and use its two-sentence response below. A student submits **only one** 1–2 sentence response; (a) and (c) are clearly identified **alternatives**:

(a) Linear density. I would study customers' spending in thousands of Colombian pesos on a promotional service priced continuously between 20 and 60 thousand pesos, proposing a linearly decreasing density over $[20, 60]$. I would check whether real observations support that proposed model.

(b) Exponential waiting times. I would study the minutes between payment requests reaching a small business during a stable trading period, proposing an exponential waiting-time model if arrivals are independent and occur at a constant average rate. I would check that assumption with real observations.

(c) Normal measurements. I would study the fill volumes in millilitres of bottles sold by a beverage producer, proposing a normal model during stable production. I would check whether real measurements support that model.

Each describes possible real-life economic data and a *proposed* model requiring validation; none claims that uncollected data follow the model. Other plausible responses are acceptable. Each example is a brief exam response, not an empirical analysis or a completed online report. The exam prints six points beside the brief selection and identifies the separate online report as the C10 evidence; no separate report point allocation is stated.

Separate online PDF report (not the brief exam response). Submit through **Open**, focusing on the **one selected option**. Investigate an economic or financial question using real data and the selected continuous model. Include simulation, sampling analysis, and portfolio-risk measures; justify methods and present results, assumptions, limitations, and evidence-based conclusions. These are the broader C10 requirements; no completed report, dataset, simulation, portfolio analysis, or empirical conclusion is supplied by these examples.



18 Summarized Solutions

Notation and conventions. $\nu = n - 1$, $c = (x - m)/a$, $x = m + ac$; $X_T = m + aT$, $T \sim t_\nu$; $X_N = m + aZ$, $Z \sim N(0, 1)$. Use the stated approximate SD directly as scale a: the models share a scale, not an actual variance. These are signed-deviation models; do not use $a/\sqrt{n}$ or interpret inverse intervals as confidence intervals for a population mean. Negative contextual values mean below the relevant reference; positive values mean above it.

All results are approximations from the supplied printed tables. No interpolation, CDF/inverse-CDF calculators or exact distribution values are used. Direct normal bounds are rounded to two decimals before lookup; inverse normal cutoffs use the closest printed CDF entry, with the 0.0500/0.9500 tie selecting $|z| = 1.64$. Retain three-decimal t and two-decimal z cutoffs. In tables marked ≈, numerical results are table-based/rounded; = denotes identities or arithmetic with selected working values.

C8

A. Direct probabilities — Problems 1–5

Problem	ν	Event	Standardized bound(s)	Table-area calculation	Probability (≈)
1	17	$8.441 < X < 14.58$	$c_L = +0.863$ $c_U = +1.740$	$0.20 - 0.05$	**0.15**
2	7	$X < -7.875$ or $X > 8.675$	$c_L = -1.895$ $c_U = +1.415$	$0.05 + 0.10$	**0.15**
3	10	$X < -5.142$	$c = -1.812$	0.05	**0.05**
4	13	$X > -12.83$	$c = -0.870$	$1 - 0.20$	**0.80**
5	8	$-9.5016 < X < -4.4004$	$c_L = -2.306$ $c_U = -0.889$	$0.20 - 0.025$	**0.175**

Event units (1–5): kWh, thousands of dollars, minutes, dollars, tonnes. Rows 1 and 5 are intervals; row 2 combines disjoint tails; rows 3 and 4 are single tails.

B. Inverse cutoffs and intervals — Problems 6–10

Problem	ν	Target probability/tail(s)	Signed t cutoff(s)	Contextual result (≈)	Units
6	9	Equal-tail limits: $P(X < L) = P(X > U) = 0.025$	$c_L = -2.262$ $c_U = +2.262$	$L = -3.67$ $U = +2.67$	g
7	15	Single lower cutoff: $P(X < x) = 0.10$	$c = -1.341$	$x = -11.55$	dollars
8	16	Unequal-tail limits: $P(X < L) = 0.01$ $P(X > U) = 0.10$	$c_L = -2.583$ $c_U = +1.337$	$L = -20.58$ $U = +6.86$	min
9	21	Single upper cutoff: $P(X > x) = 0.025$	$c = +2.080$	$x = +19.14$	min
10	12	Central interval: $P(L < X < U) = 0.90$ $P(X < L) = P(X > U) = 0.05$	$c_L = -1.782$ $c_U = +1.782$	$L = -12.82$ $U = +22.82$	tonnes

L = lower endpoint; U = upper endpoint. Problem 8: outside = $0.01 + 0.10 = 0.11$; inside = $1 - 0.11 = 0.89$ (selected table areas).

C9

Comparison measures. $E_A = |r_T - r_N|$, $E_{Pt} = 100E_A/|r_T|$, $E_{Pn} = 100E_A/|r_N|$. For direct problems, $r_T = p_T$, $r_N = p_N$; for inverse problems, r_T, r_N are contextual cutoffs after back-transformation. These compare model results, not errors against observed truth. E_{Pt}, E_{Pn} are relative percentages; a probability difference of E_A corresponds to $100E_A$ percentage points.

A. Direct model comparisons — Problems 11–13

Problem/event	ν	p_T	p_N	E_A	E_{Pt}	E_{Pn}
11: $X > -0.7285$ (min)	12	0.8000	0.8078	0.0078	0.98%	0.97%
12: $-8.25 < X < 16.32$ (litres)	13	0.8750	0.8961	0.0211	2.41%	2.35%
13: $X < -13.372$ (thousands of dollars)	10	0.0500	0.0351	0.0149	29.80%	42.45%

All numerical results $\approx$; p_T, p_N, E_A are unitless. Bounds: 11: $c = -0.873 \to z = -0.87$; 12: $c_L = -1.350 \to z_L = -1.35$, $c_U = +2.160 \to z_U = +2.16$; 13: $c = -1.812 \to z = -1.81$.

B. Inverse model comparisons — Problems 14–16

Problem/ endpoint	Target	r_T	r_N	E_A	E_{Pt}	E_{Pn}	Units
14 / L	$P(X < L) = 0.025$	−9.34	−8.00	1.34	14.35%	16.75%	kg
14 / U	$P(X > U) = 0.025$	12.94	11.60	1.34	10.36%	11.55%	kg
15 / x	$P(X < x) = 0.05$	−6.88	−6.22	0.66	9.59%	10.61%	dollars
16 / x	$P(X > x) = 0.10$	7.73	7.52	0.21	2.74%	2.82%	min

All numerical results $\approx$; r_T, r_N, E_A have the stated units. Problem 14 central 95% intervals: $t \approx (-9.34, 12.94)$ kg; normal $\approx (-8.00, 11.60)$ kg. The t interval extends 1.34 kg farther from $m = 1.8$ at both ends.

Cutoffs: 14 ($\nu = 10$): $c_{T,L} = -2.228$, $c_{T,U} = +2.228$, $z_L = -1.96$, $z_U = +1.96$; 15 ($\nu = 8$): $c_T = -1.860$, $z = -1.64$; 16 ($\nu = 17$): $c_T = +1.333$, $z = +1.28$. Use unrounded back-transformed cutoffs for comparisons; for 16, $r_T = 7.7320$, $r_N = 7.52$, $E_A = 0.2120$ in selected working values, not the displayed 7.73.

C10

Component	Summarized response or requirement
Selected option	**(b), exponential waiting times.**
Brief exam response	I would study the minutes between payment requests reaching a small business during a stable trading period, proposing an exponential waiting-time model if arrivals are independent and occur at a constant average rate. I would check that assumption with real observations.
Alternatives	**(a)**, linear density over a finite interval; **(c)**, normal measurements. Acceptable alternatives; students submit **only one 1–2 sentence response**. Other plausible responses are acceptable.
Separate online report	**PDF submitted through Open**, using real economic or financial data and the **one selected model**. Include simulation, sampling analysis, portfolio-risk measures, justified methods, results, assumptions, limitations, and evidence-based conclusions.

The brief response proposes data and a model to validate; it supplies no completed report or empirical findings. The online report is the C10 evidence; no separate report point allocation is stated.